

2016.0 RANGE ROVER (LG), 501-25

BODY REPAIRS – GENERAL INFORMATION

DESCRIPTION AND OPERATION

INTRODUCTION

The body plays a significant role in the increasing trend of ever more rapidly changing model variants. The different customer groups are strongly influenced by the design and shape of the body. At the same time the stability of the body plays the most important part in ensuring passenger and driver safety. Lightweight construction, alternative materials, composite materials, plastics and appropriate joining processes are all design features that characterize modern Land Rover vehicle bodies.

In terms of manufacturing technology, modern safety cell bodies can be produced almost without any problems. Land Rover guarantee high quality standards by ensuring that mechanical strength properties are tried and tested in numerous computer simulations, crash tests, by testing materials and by employing sophisticated manufacturing technologies. In the event of repairs it is vital that the production quality standards are upheld. This requires a well-equipped workshop, and places particular emphasis on the qualifications of the workshop technicians. Up-to-date knowledge of current manufacturing technologies and continuous training on new repair methods and techniques are vital for high-quality body repairs. The model-specific repair manuals and the general repair techniques provide valuable support when undertaking body repairs.

Always follow the repair instructions published in this manual. Failure to observe this instruction can result in serious impairment of vehicle safety. All specified safety requirements must be met after the work has been carried out.

C O N T E N T S

This document includes technical data and information for the all new Range Rover L405 aluminum range. The information contained within is valid at the time of production and incorporates:

- Who should use this information?
- Category A and B definition
- Accident damage and diagnosis
- Impact effects on the body
- Planning a repair
- Estimating
- Identifying the correct Land Rover Approved Body Centres
- Obtaining spare parts
- Equipment
- aluminum Information

In line with Land Rover continuous improvement program, information and data contained will be updated periodically.

All activities described within are based on the use of genuine Land Rover Parts, tools and approved materials.

W H O S H O U L D U S E T H I S I N F O R M A T I O N ?

The information and repair methods listed are designed as an aid for Land Rover Approved Body Centres achieving the Land Rover approved

Bodyshop Operating Standards. The relevant section should be read completely before commencing any repairs. Only technicians who have successfully completed the approved L405 aluminum training program should work on the model. Land Rover Approved Bodyshop Operating Standards require that the skills of technicians be regularly assessed and that any training needs identified are addressed within a reasonable time. This information complements the Land Rover training program.

The authorized repair network is divided into category A and B. The following describes the process required to identify category A or B damage.

DEFINITION OF CATEGORY A AND B

Category A (Structural)

NOTE:

Specific equipment and facilities are required to carry out Category A repairs, see the equipment section of this manual.

Damage that requires panel(s) to be replaced with any one or combination of the following procedures:

- Welded panel
- Bonded panel
- Panel secured with fixings

The following list identifies Category A procedures:

Category A procedures

- Front side member section

For additional information, refer to: [Front Side Member Section](#) (501-27

Front End Sheet Metal Repairs, Removal and Installation).

- Front side member closing panel section

For additional information, refer to: Front Side Member Closing Panel Section (501-27 Front End Sheet Metal Repairs, Removal and Installation).

- Fender apron panel

For additional information, refer to: [Fender Apron Panel](#) (501-27 Front End Sheet Metal Repairs, Removal and Installation).

- Fender apron panel closing panel

For additional information, refer to: [Fender Apron Panel Closing Panel](#) (501-27 Front End Sheet Metal Repairs, Removal and Installation).

- Fender apron panel inner reinforcement

For additional information, refer to: Fender Apron Panel Inner Reinforcement (501-27 Front End Sheet Metal Repairs, Removal and Installation).

- Suspension top mount

For additional information, refer to: [Suspension Top Mount](#) (501-27 Front End Sheet Metal Repairs, Removal and Installation).

- Front side member and suspension top mount assembly

For additional information, refer to: [Front Side Member and Suspension Top Mount Assembly](#) (501-27 Front End Sheet Metal Repairs, Removal and Installation).

- Front fender mounting panel

For additional information, refer to: [Front Fender Mounting Panel](#) (501-27 Front End Sheet Metal Repairs, Removal and Installation).

- Front fender supporting panel

For additional information, refer to: Front Fender Supporting Panel (501-27 Front End Sheet Metal Repairs, Removal and Installation).

- Front fender lower mounting bracket

For additional information, refer to: [Front Fender Lower Mounting Bracket](#) (501-27 Front End Sheet Metal Repairs, Removal and Installation).

- Hood latch panel

For additional information, refer to: [Hood Latch Panel](#) (501-27 Front End Sheet Metal Repairs, Removal and Installation).

- Hood latch panel mounting bracket

For additional information, refer to: [Hood Latch Panel Mounting Bracket](#) (501-27 Front End Sheet Metal Repairs, Removal and Installation).

- Roof panel

For additional information, refer to: [Roof Panel](#) (501-28 Roof Sheet Metal Repairs, Removal and Installation).

- Roof front panel

For additional information, refer to: [Roof Front Panel](#) (501-28 Roof Sheet Metal Repairs, Removal and Installation).

- Roof rear panel

For additional information, refer to: [Roof Rear Panel - Standard Wheel Base](#) (501-28 Roof Sheet Metal Repairs, Removal and Installation).

- Side panel

For additional information, refer to: [Side Panel - Standard Wheel Base](#) (501-29 Side Panel Sheet Metal Repairs, Removal and Installation).

- Side panel front section

For additional information, refer to: [Side Panel Front Section](#) (501-29 Side Panel Sheet Metal Repairs, Removal and Installation).

- A-pillar reinforcement

For additional information, refer to: ['A' Pillar Reinforcement](#) (501-29 Side Panel Sheet Metal Repairs, Removal and Installation).

- Rocker panel

For additional information, refer to: [Rocker Panel](#) (501-29 Side Panel Sheet Metal Repairs, Removal and Installation).

- Rocker panel inner reinforcement

For additional information, refer to: [Rocker Panel Inner Reinforcement - Standard Wheel Base](#) (501-29 Side Panel Sheet Metal Repairs, Removal and Installation).

- Rocker panel section

For additional information, refer to: [Rocker Panel Section](#) (501-29 Side Panel Sheet Metal Repairs, Removal and Installation).

- Rocker panel and B-pillar inner

For additional information, refer to: [Rocker Panel and B Pillar Inner](#) (501-29 Side Panel Sheet Metal Repairs, Removal and Installation).

- B-pillar outer panel

For additional information, refer to: ['B' Pillar Outer Panel](#) (501-29 Side Panel Sheet Metal Repairs, Removal and Installation).

- B-pillar reinforcement

For additional information, refer to: [B-Pillar Reinforcement](#) (501-29 Side Panel Sheet Metal Repairs, Removal and Installation).

- Quarter panel

For additional information, refer to: [Quarter Panel](#) (501-30 Rear End Sheet Metal Repairs, Removal and Installation).

- Quarter panel reinforcement

For additional information, refer to: [Quarter Panel Reinforcement](#) (501-30 Rear End Sheet Metal Repairs, Removal and Installation).

- Inner quarter Panel

For additional information, refer to: Inner Quarter Panel Assembly (501-30 Rear End Sheet Metal Repairs, Removal and Installation).

- Inner quarter Panel Lower

For additional information, refer to: [Inner Quarter Panel Lower](#) (501-30 Rear End Sheet Metal Repairs, Removal and Installation).

- Inner quarter panel upper

For additional information, refer to: [Inner Quarter Panel Upper - Standard Wheel Base, Vehicles With: Glass Roof Panel](#) (501-30 Rear End Sheet Metal Repairs, Removal and Installation).

- Inner quarter Panel Extension

For additional information, refer to: Inner Quarter Panel Extension (501-30 Rear End Sheet Metal Repairs, Removal and Installation).

- Rear crossmember

For additional information, refer to: [Rear Crossmember \(501-30 Rear End Sheet Metal Repairs, Removal and Installation\)](#).

- Rear side member section

For additional information, refer to: [Rear Side Member Section \(501-30 Rear End Sheet Metal Repairs, Removal and Installation\)](#).

- Back panel outer

For additional information, refer to: [Back Panel Outer \(501-30 Rear End Sheet Metal Repairs, Removal and Installation\)](#).

- Back panel inner

For additional information, refer to: [Back Panel Outer \(501-30 Rear End Sheet Metal Repairs, Removal and Installation\)](#).

- Rear floor panel section

For additional information, refer to: [Rear Floor Panel Section \(501-30 Rear End Sheet Metal Repairs, Removal and Installation\)](#).

- Rear floor side extension

For additional information, refer to: [Rear Floor Side Extension \(501-30 Rear End Sheet Metal Repairs, Removal and Installation\)](#).

- Rear lamp panel inner

For additional information, refer to: [Rear Lamp Panel Inner \(501-30 Rear End Sheet Metal Repairs, Removal and Installation\)](#).

- Rear lamp panel outer

For additional information, refer to: [Rear Lamp Panel Outer \(501-30 Rear End Sheet Metal Repairs, Removal and Installation\)](#).

- Rear wheelhouse outer

For additional information, refer to: [Rear Wheelhouse Outer \(501-30 Rear End Sheet Metal Repairs, Removal and Installation\)](#).

- Water drain panel lower

For additional information, refer to: [Water Drain Panel Lower \(501-30 Rear End Sheet Metal Repairs, Removal and Installation\)](#).

Category B (Cosmetic)

Damage associated to bolt on panel(s) that are replaced, or light surface damage to exterior cosmetic panels.

The following list identifies Category B procedures:

- Front bumper
For additional information, refer to: [Front Bumper](#) (501-19 Bumpers, Removal and Installation).
- Front bumper cover
For additional information, refer to: [Front Bumper Cover](#) (501-19 Bumpers, Removal and Installation).
- Front bumper lower cover
For additional information, refer to: [Front Bumper Lower Cover](#) (501-19 Bumpers, Removal and Installation).
- Rear bumper
For additional information, refer to: [Front Bumper](#) (501-19 Bumpers, Removal and Installation).
- Rear bumper cover
For additional information, refer to: [Front Bumper Cover](#) (501-19 Bumpers, Removal and Installation).
- Cowl panel grille
For additional information, refer to: [Cowl Panel Grille](#) (501-02 Front End Body Panels, Removal and Installation).
- Engine undershield
For additional information, refer to: [Engine Undershield](#) (501-02 Front End Body Panels, Removal and Installation).
- Fender
For additional information, refer to: [Fender](#) (501-02 Front End Body Panels, Removal and Installation).
- Fender splash shield
For additional information, refer to: [Fender Splash Shield](#) (501-02 Front End Body Panels, Removal and Installation).

- Hood latch panel

For additional information, refer to: [Hood Latch Panel](#) (501-27 Front End Sheet Metal Repairs, Removal and Installation).

- Front door

For additional information, refer to: [Front Door](#) (501-03 Body Closures, Removal and Installation).

- Fuel filler door assembly

For additional information, refer to: [Fuel Filler Door Assembly](#) (501-03 Body Closures, Removal and Installation).

- Fuel filler door

For additional information, refer to: [Fuel Filler Door](#) (501-03 Body Closures, Removal and Installation).

- Liftgate

For additional information, refer to: [Liftgate](#) (501-03 Body Closures, Removal and Installation).

- Rear door

For additional information, refer to: [Rear Door - Standard Wheel Base](#) (501-03 Body Closures, Removal and Installation).

ACCIDENT DAMAGE AND DIAGNOSIS

General notes:

- Exact diagnosis of the extent of the damage enables proper repair planning
- All body repairs must be carried out in accordance with the guidelines in this Body Repair Manual
- The stability and strength properties of the body must be taken into account during body repairs. The body has exact defined deformation patterns that must not be affected by any repair work
- For instance, the crumple zones absorb the bulk of the impact energy. If any unprofessional repair techniques or methods are used in these areas

then this can pose a fundamental threat to vehicle safety

Hidden damage:

- As well as looking at external indicators like flaked off paint, it is vital to check for hidden body damage or deformation that is not visible from the outside. Large attached parts like bumpers and inner fenders often need to be removed to allow accurate assessment of damage to underlying body parts
- Gap dimensions offer another alternative for diagnosis by visual inspection. If any changes or misaligned edges are apparent, then this usually indicates that the dimensions of the affected part are incorrect

IMPACT EFFECTS ON THE BODY

It is possible to deduce the overall extent of damage from the direction and magnitude of the impact forces. This does however require extensive body-specific knowledge.

If, for instance, an impact occurs at the front left-hand side member, then the right-hand side member is usually also affected as a result of the rigid body-shell design (crossmember). Often the length of this side member will not have changed, but because of the rigid body-shell design it may have moved from its original position (often only by a very small amount). If any deviations are present this can usually be detected by checking the gap dimensions between door and fender or by checking for changes in dimension.

In the case of more severe impacts, the front part of the body cannot absorb all of the impact energy, and the passenger cell is also deformed. Here the impact energy is transferred via the side member to the A-pillar. This results in deformations in the area of the roof and the door rocker panel.

PLANNING A REPAIR

The following decisions have to be made before the repairs are started

- Does the vehicle need to be put on a body repair jig?
- Does the body need to be measured?
- Do aggregates like engine or axles need to be removed?
- Which body parts need to be renewed?
- Which body parts can be repaired?

BATTERY CARE STANDARDS

To ensure that the correct maintenance actions are carried out before and during a repair.

For additional information, refer to: Battery Care Requirements (414-00 Battery and Charging System - General Information, Description and Operation).

HYBRID ELECTRIC VEHICLES (HEV)

When planning a repair on Hybrid Electric Vehicle (HEV) make sure that the vehicle is inspected by an authorised person. This will then determine whether the vehicle needs to be decommissioned before any repair work can commence.

If the vehicle needs to be decommissioned this can only be carried out by an authorised person. The authorised person will also issue a permit to work on the vehicle.

ESTIMATING

The authorized repair network is divided into category A and B. This section describes the process required to identify category A or B damage. Should damage be identified as category A, the vehicle should be referred to the nearest category A facility.

In line with the Land Rover continuous improvement program, information and data contained in this manual will be updated through Land Rover technical bulletins.

THE ESTIMATING PROCESS

Step 1:

Visual inspection, assess overall vehicle condition. Has damage travelled, distorting internal structural panels?

This is a visual check of the complete vehicle.

Possible indicators:

- Panel misalignment
- Panel gaps no longer uniform
- Hinged panels catch on locks
- Exterior cosmetic panels show signs of stress distortion
- Customer describes unusual driving characteristics post incident

If structural distortion is visible, vehicle body alignment should be checked/corrected on a body repair jig, refer to category A facility.

If no visual indications of damage transfer, go to Step 2.

Step 2:

If no visual signs indicate distortion to internal structural panels, but the customer is concerned with unusual driving characteristics post incident, and there is no damage to the suspension. Vehicle body alignment should be checked using four-wheel alignment geometry equipment.

If structural distortion is identified refer to category A facility.

Step 3:

If there is no sign or concern regarding vehicle body alignment, focus on the local area of damage.

Do any of the panels require replacement?

If no, go to Step 4.

If yes, are any of the panels secured with:

- Bonding adhesive?
- Fixings?
- Weld?

If yes, refer to category A facility.

Step 4:

Do any of the panels require welding i.e. ripped or torn aluminum panel?

If no, go to Step 5.

If yes, refer to category A facility.

Step 5:

Can the damage be repaired by hand using serrated spoon, dolly, file and/or hot air?

If no, refer to category A facility.

IDENTIFYING THE CORRECT LAND ROVER APPROVED BODY CENTRES

Through extensive research, Land Rover has identified the most appropriate method to rectify aftermarket damage. It is key to safety and vehicle integrity that only the approved methods, materials and equipment are used.

Issues that could arise from the use of non-approved methods, materials and equipment include:

- Warranty invalidation
- Safety
- Galvanic corrosion

Therefore, category A damage should always be referred to a category A Land Rover Approved Body Centres.

OBTAINING SPARE PARTS

The availability of spare parts often determines how easily the body repairs can be carried out. The following procedure is recommended:

- Obtain all the data for the vehicle, including type, vehicle identification number, trim code, engine identification letters, initial registration etc.
- Establish all of the metal parts that need to be renewed
- Establish all of the attached parts that need to be renewed, including small parts like rivets, clips etc.

EQUIPMENT

Land Rover Approved Body Centres are expected to be equipped in line with the Body Centre Tools and Equipment standards.

The approved body equipment for the all new L405 aluminum range is listed as either:

- Category A
- Category B

Body centre approved equipment is also available to order on a global basis from SPX Service Solutions:

COUNTRY	NAME	PHONE	FAX
UK	Christine Gibson	+44 1327 303 400	+ 44 1327 303 499
France	Sophie Leger	+33 2 52 84 00 40	+ 33 2 43 60 43 62
Spain	Azucena Granado	+34 949 208 329	+34 949 208 327
Non EU Markets	Oliver Lehnasch	+49 6182 959 345	+49 6182 959 245
Benelux	Sonja Visschers	+31 455 678 870	+31 455 678 871
Italy	Sonia Ianelli	+39 0521 338 170	+39 0521 837 370
All other EU Markets	Theodor Gardocki	+49 6182 959 375	+49 6182 959 246

Nordic	Sirpa Kulmala	+49 6182 959 495	+49 6182 959 245
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A copy of the approved equipment standards for category 'A' and 'B' repair facilities is available on request from and the Global Body Centre Standards have been published on the HUB, along with the personal protective equipment requirements, at the following locations:

<https://jaguarhub.hub.tagworldwide.com/index.cfm?brand=42701>

<https://landroverhub.hub.tagworldwide.com/index.cfm?brand=42701>

ALUMINUM INFORMATION

The design of the modern motor vehicle attempts to overcome two conflicting needs:

- Fuel economy - lighter, aerodynamic and fuel sensitive technology
- High levels of comfort - this often equates to higher specifications and more accessories

aluminum alloy is the ideal material to meet these demands; it provides a lighter vehicle body with improved rigidity. aluminum is different from traditional Steel, with the correct knowledge and suitable tools it is easily repaired.

There are two aluminum alloys discussed in this manual, the attributes detailed in the following table should be considered when deciding to repair or replace:

	6XXX	5754
Material Description	6xxx Series is a Magnesium/Silicon aluminum alloy	5000 Series is an aluminum alloy with Magnesium content
Location on Vehicle	This alloy is used mainly in the outer body panels. Thickness: 0.9 - 2.0mm	Internal structural panels Thickness: 1.0 - 3.0mm
Attributes	High dent resistance	Strength and durability

Reparability	Yes - Light damage only	NOTE: All repairs to 5754 are restricted to Category A Land Rover approved Body Repair Centres only. Yes - Light damage only. Limited straightening is acceptable
Heat During Repair	CAUTION: material sensitive to heat. Maximum temperature range: 140°C.	Yes - must be heated to maintain original alloy properties. Maximum Temperature range: 250°C.
Notes	WARNING: DO NOT USE AN OPEN FLAME e.g. Oxygen Acetylene Equipment. Equipment: Hot air gun. Use panel temperature indicator strips. Heat crayons and digital thermometers to identify temperature.	WARNING: DO NOT USE AN OPEN FLAME e.g. Oxygen Acetylene Equipment. Equipment: Induction heater. Use panel temperature indicator strips. Heat crayons and digital thermometers to identify temperature. Heat must be applied and maintained during the straightening process.

Other materials are used in the construction of the all new L405 aluminum range, however, repair of these materials is not covered as they fall outside the scope of this manual.

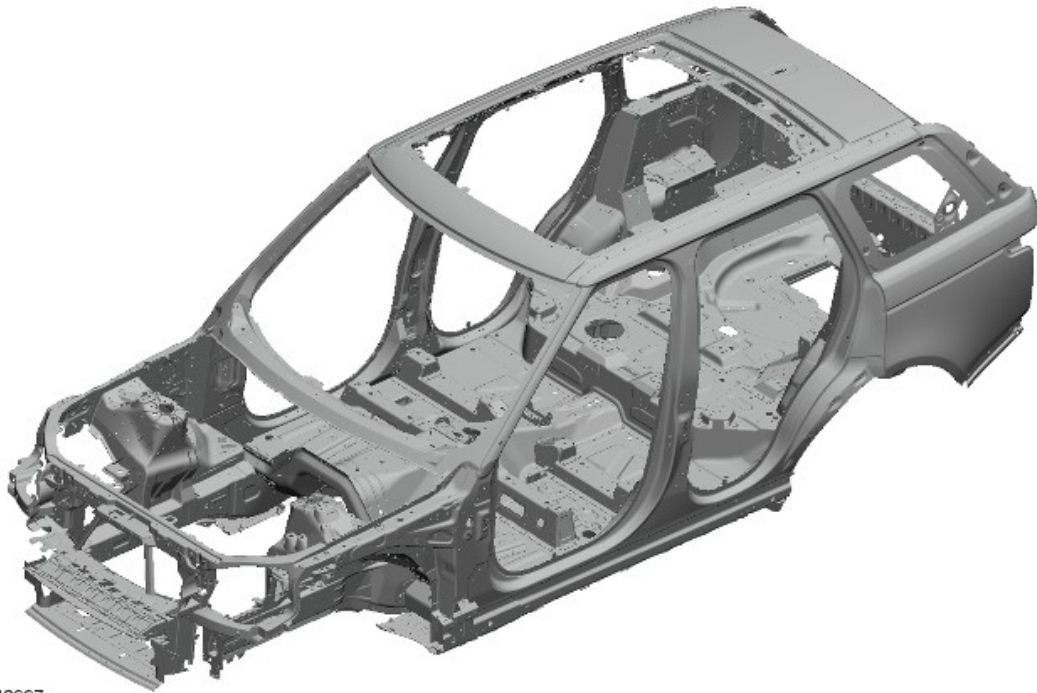
Steel, (including Bake Hardened (BH) and High Strength Low Alloy (HSLA): Used in seat belt anchorages, hinge reinforcements and various small brackets and mountings.

Magnesium Die Cast Alloy (AM60B): Used in the hood latch panel.

Plastics: Used in the front fender support bracket and for reinforcements in the B-pillar and rear side member.

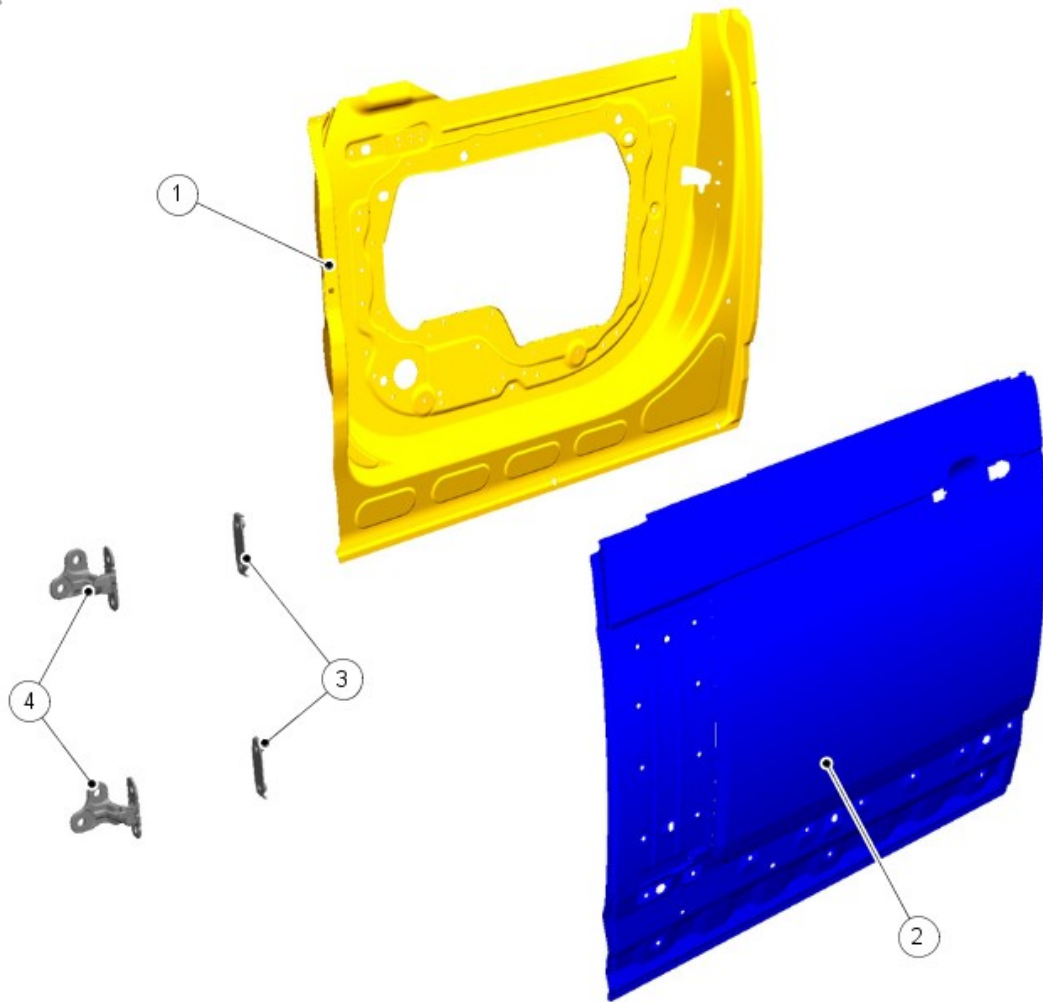
The following illustrations identify the aluminum alloys and other materials used in body construction.

Bodyshell



E142997

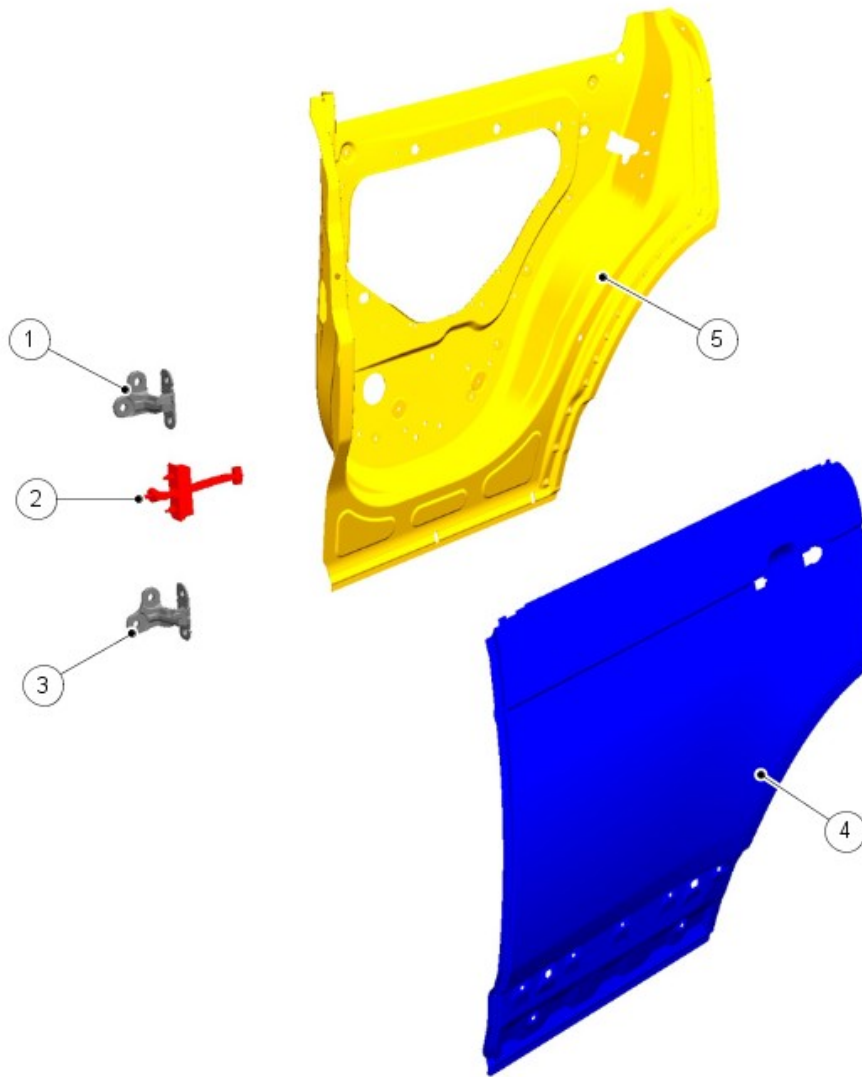
Body closures - front door



E144632

ITEM	MATERIAL CODE	MATERIAL DESCRIPTION
1	5182	aluminum
2	6111T4	aluminum
3	-	Steel
4	-	Steel

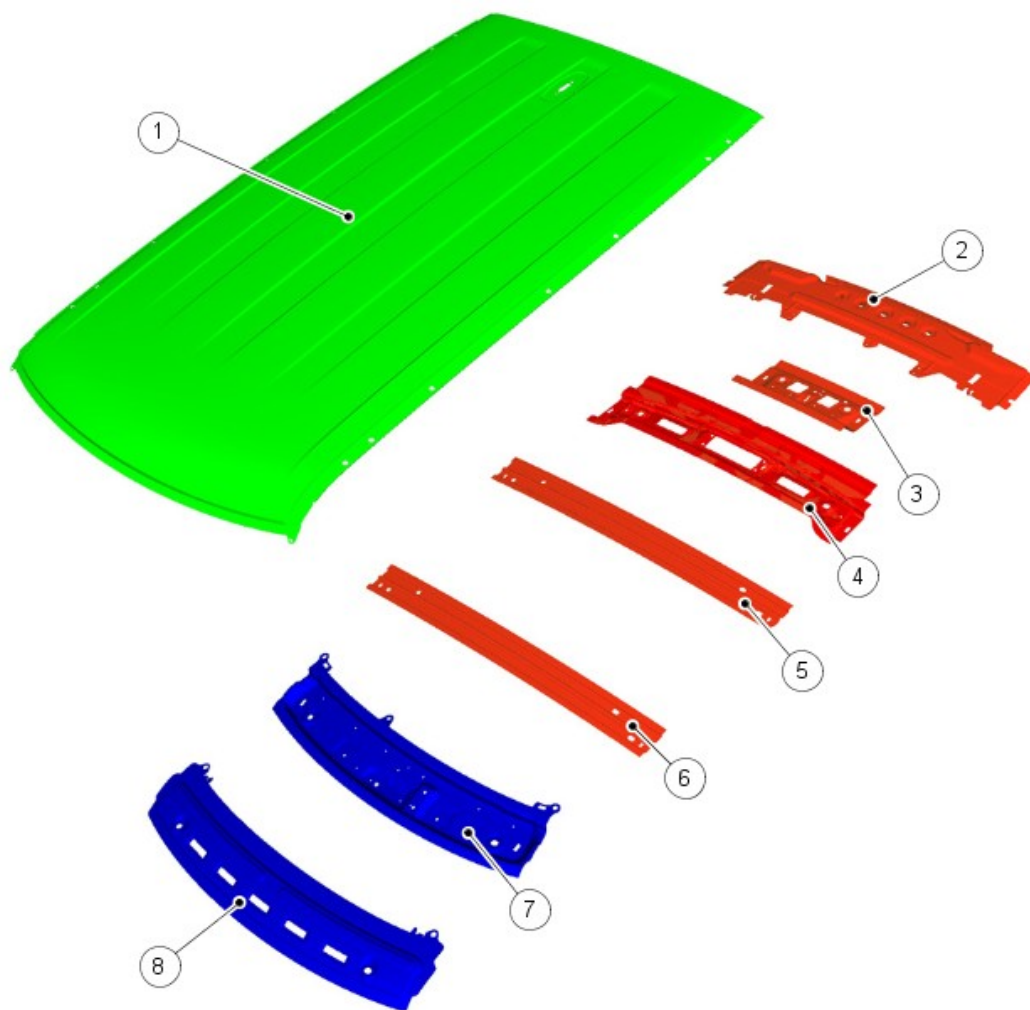
Body closures - rear door



E144633

ITEM	MATERIAL CODE	MATERIAL DESCRIPTION
1	-	Steel
2	5754NG	aluminum
3	-	Steel
4	6111T4	aluminum
5	5182	aluminum

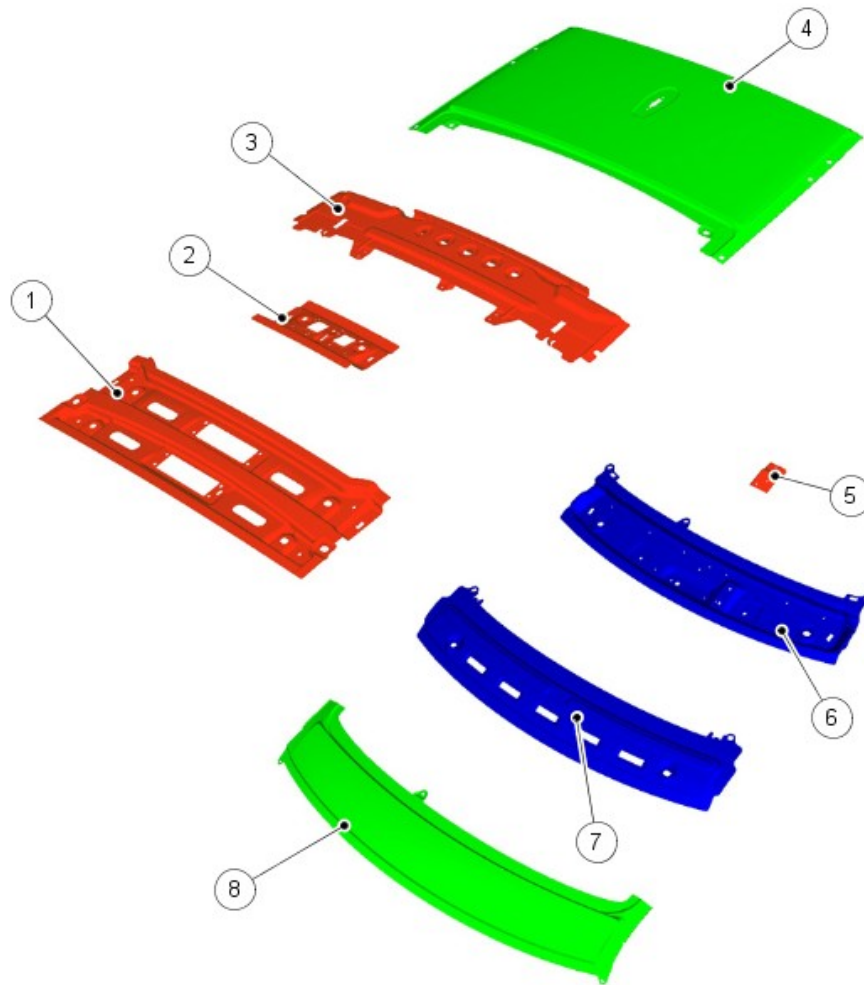
Roof panels



E143702

ITEM	MATERIAL CODE	MATERIAL DESCRIPTION
1	AC600	aluminum
2	5754NG	aluminum
3	5754NG	aluminum
4	5754NG	aluminum
5	5754NG	aluminum
6	5754NG	aluminum
7	6111T4	aluminum
8	6111T4	aluminum

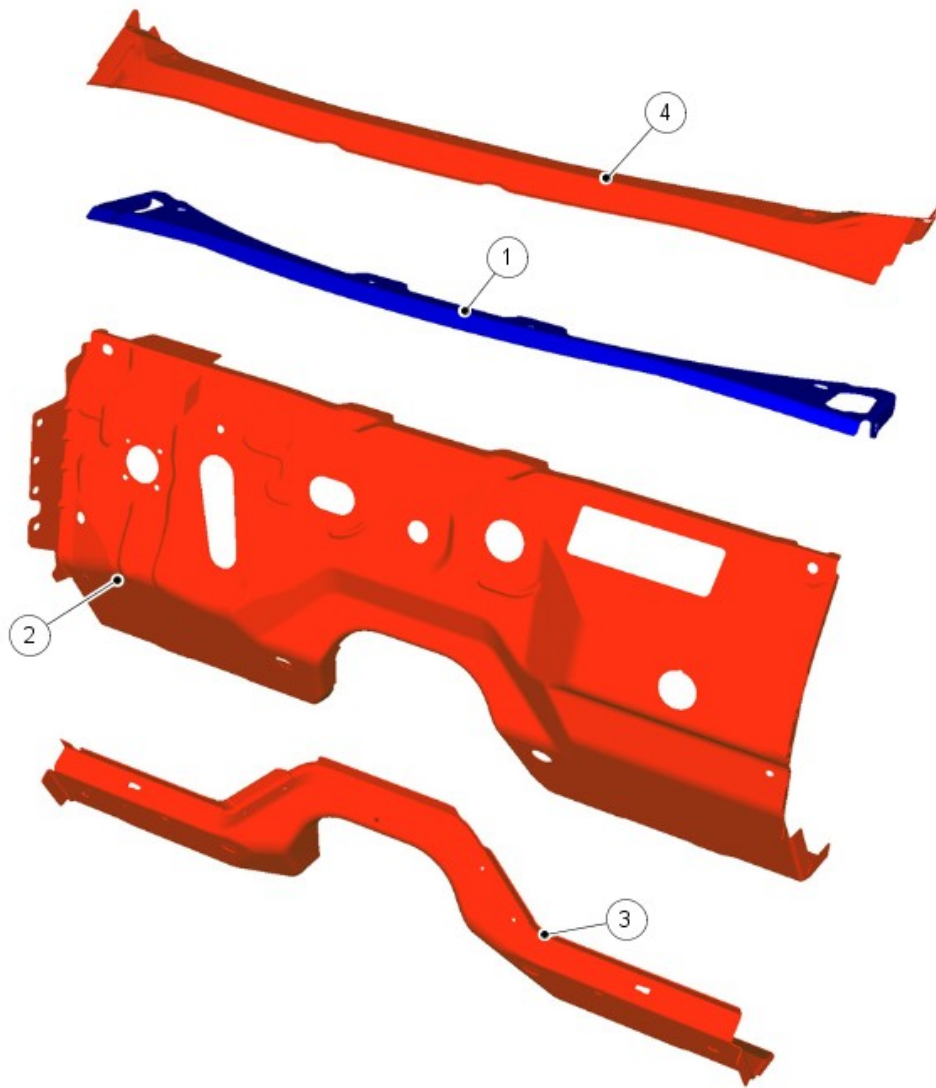
Roof panels (moonroof)



E143007

ITEM	MATERIAL CODE	MATERIAL DESCRIPTION
1	5754NG	aluminum
2	5754NG	aluminum
3	5754NG	aluminum
4	AC600	aluminum
5	5754NG	aluminum
6	6111T4	aluminum
7	6111T4	aluminum
8	AC600	aluminum

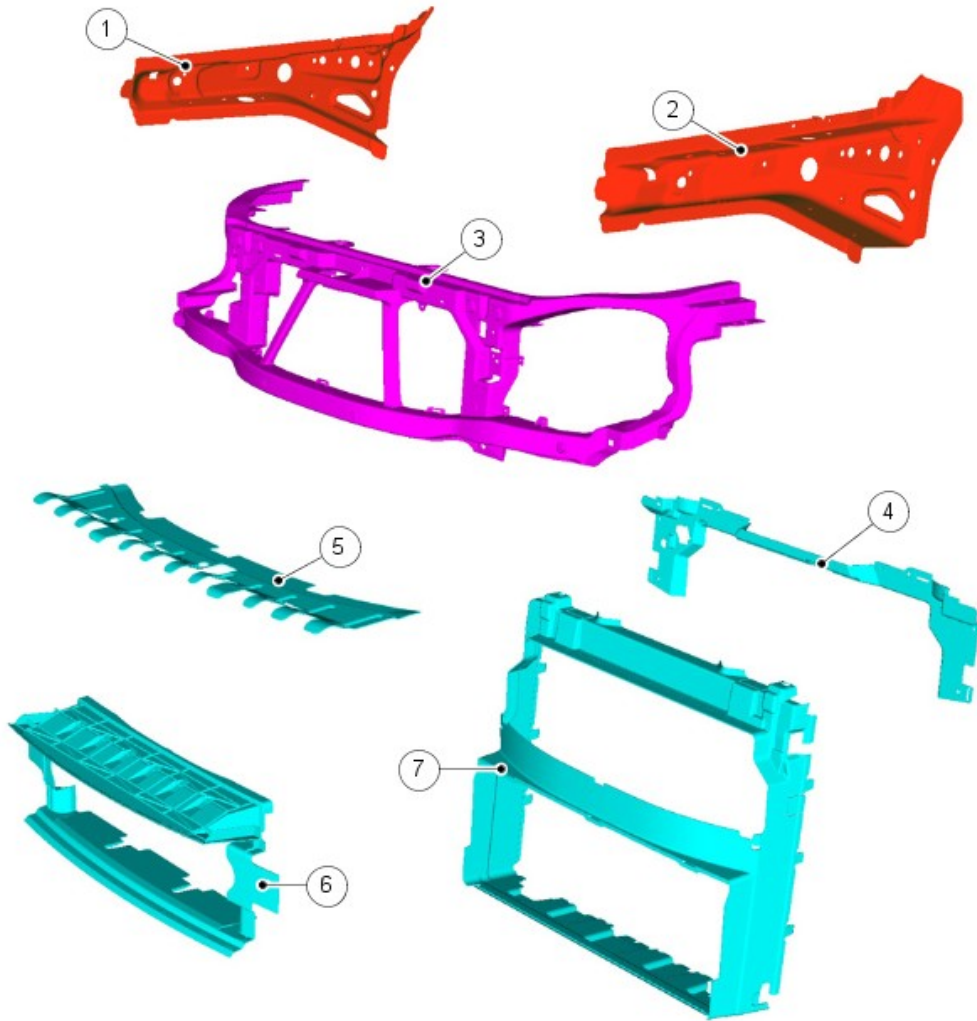
Front end panels



E144634

ITEM	MATERIAL CODE	MATERIAL DESCRIPTION
1	6111T4	aluminum
2	5754NG	aluminum
3	5754NG	aluminum
4	5754NG	aluminum

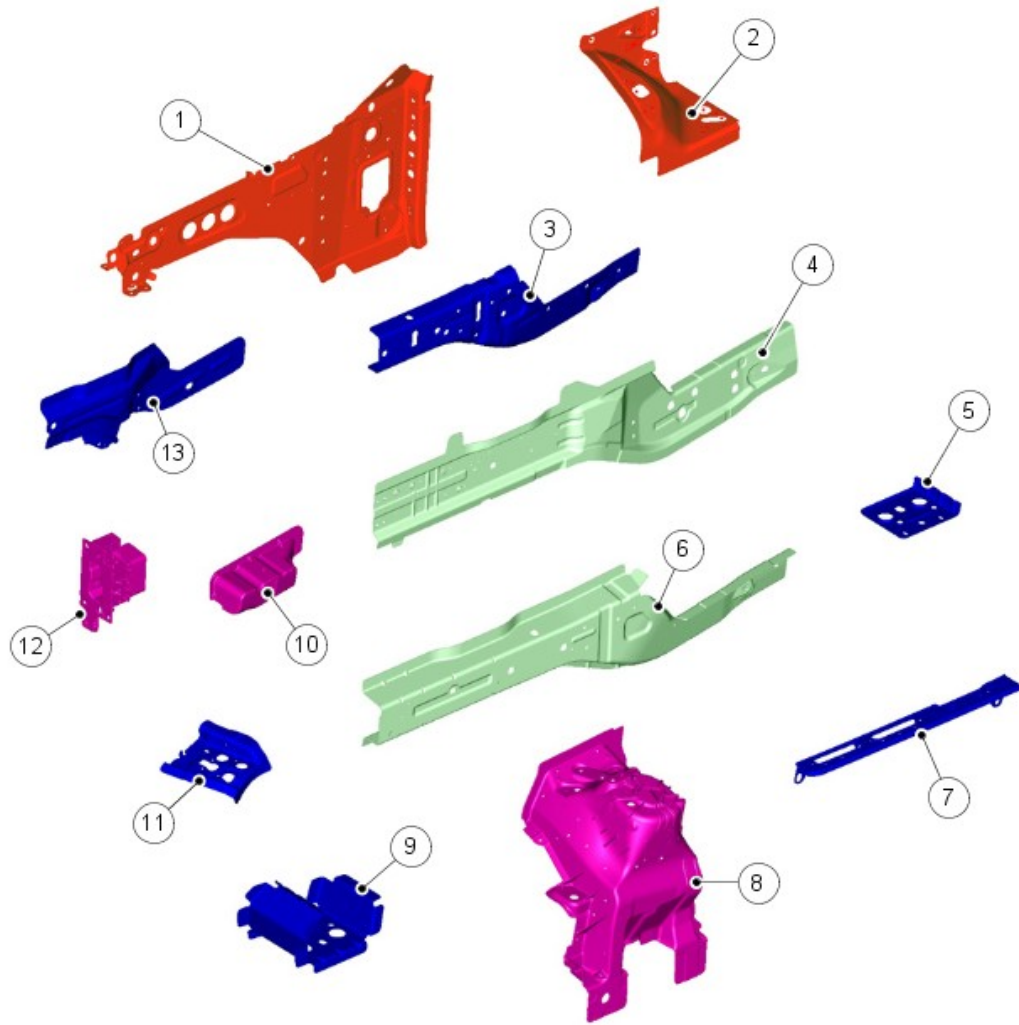
Front end panels - continued



E144635

ITEM	MATERIAL CODE	MATERIAL DESCRIPTION
1	5754NG	aluminum
2	5754NG	aluminum
3	ES-AW93-8B040	MAGNESIUM DIE CAST (AM60B)
4	-	Thermoplastic
5	-	Thermoplastic
6	-	Thermoplastic
7	-	Thermoplastic

Front end panels - continued

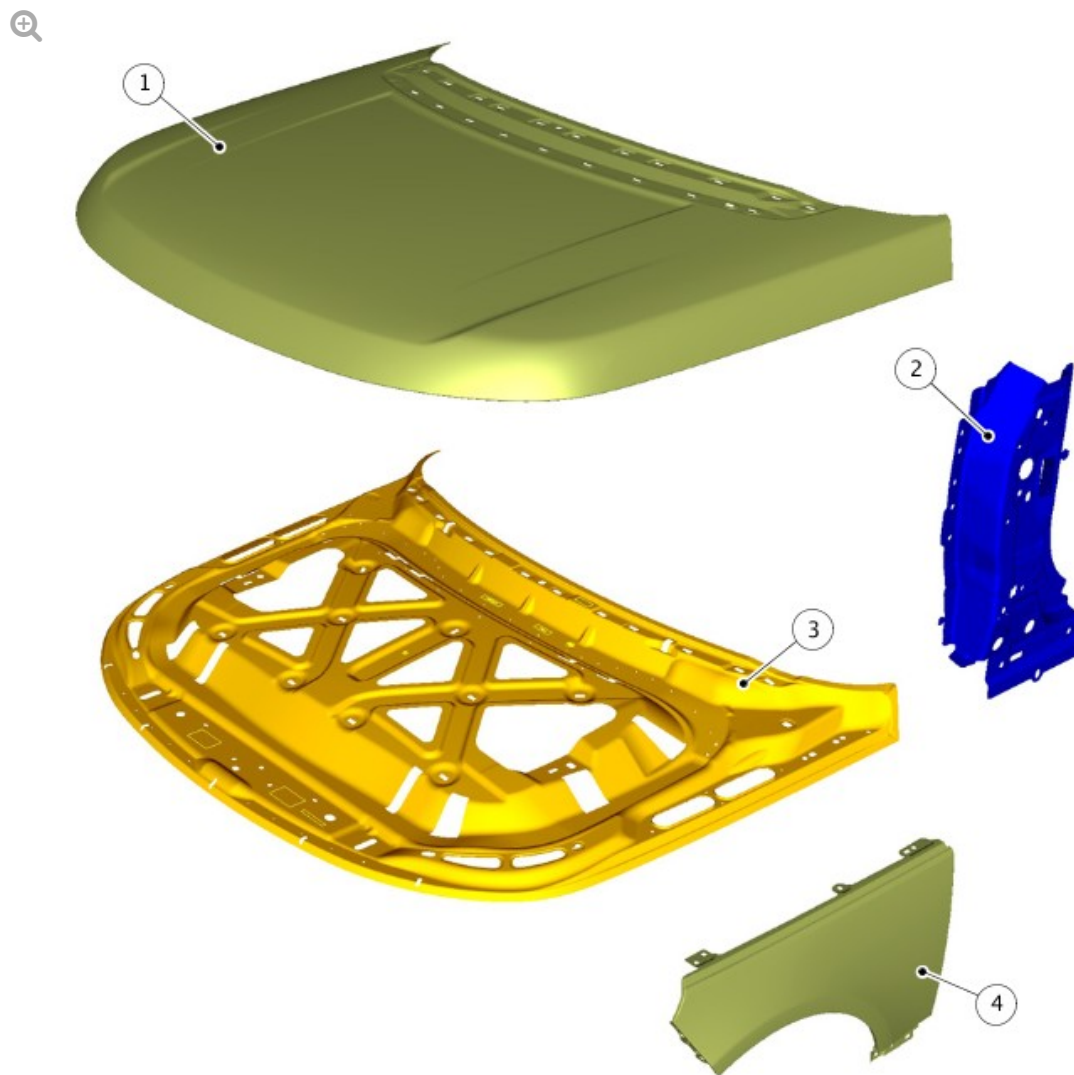


E144636

ITEM	MATERIAL CODE	MATERIAL DESCRIPTION
1	5754NG	aluminum
2	5754NG	aluminum
3	6111T4	aluminum
4	AC300 T61	aluminum
5	6111T4	aluminum
6	AC300 T61	aluminum
7	6111T4	aluminum
8	ADCHT	aluminum
9	6111T4	aluminum
10	6111T4	aluminum

11	ADCHT	aluminum
12	ADCHT	aluminum
13	6111T4	aluminum

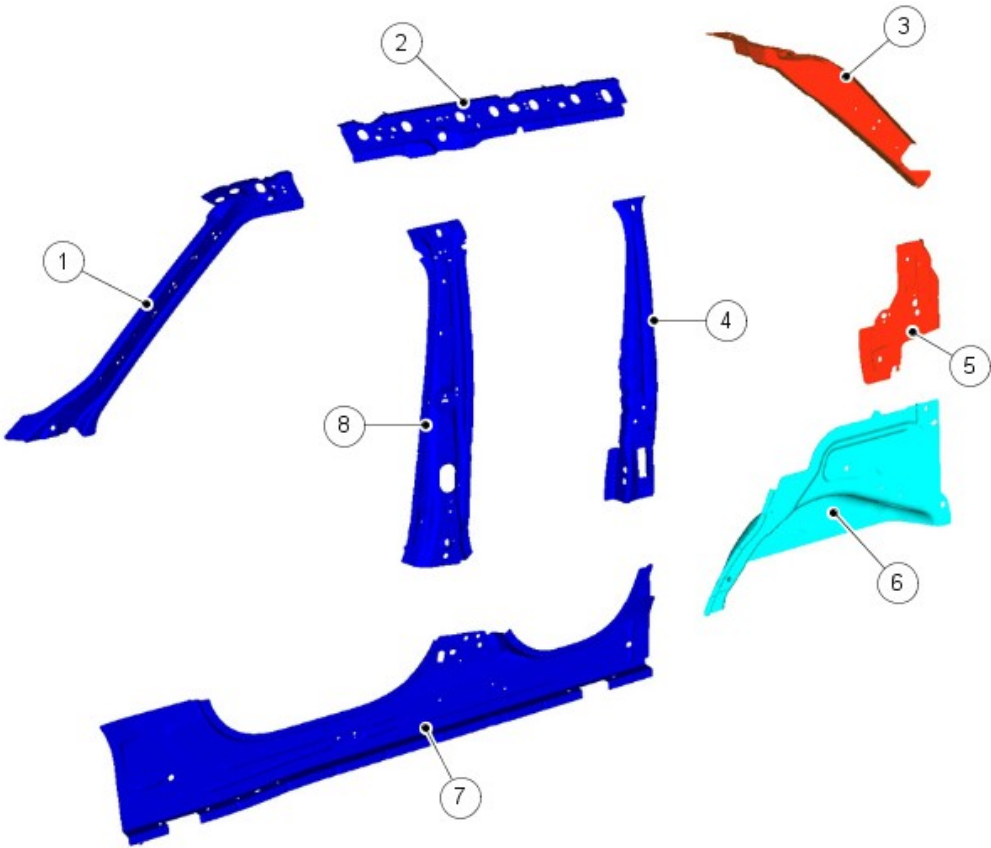
Front end panels - continued



E144637

ITEM	MATERIAL CODE	MATERIAL DESCRIPTION
1	AC170PX	aluminum
2	6111T4	aluminum
3	5182	aluminum
4	AC170PX	aluminum

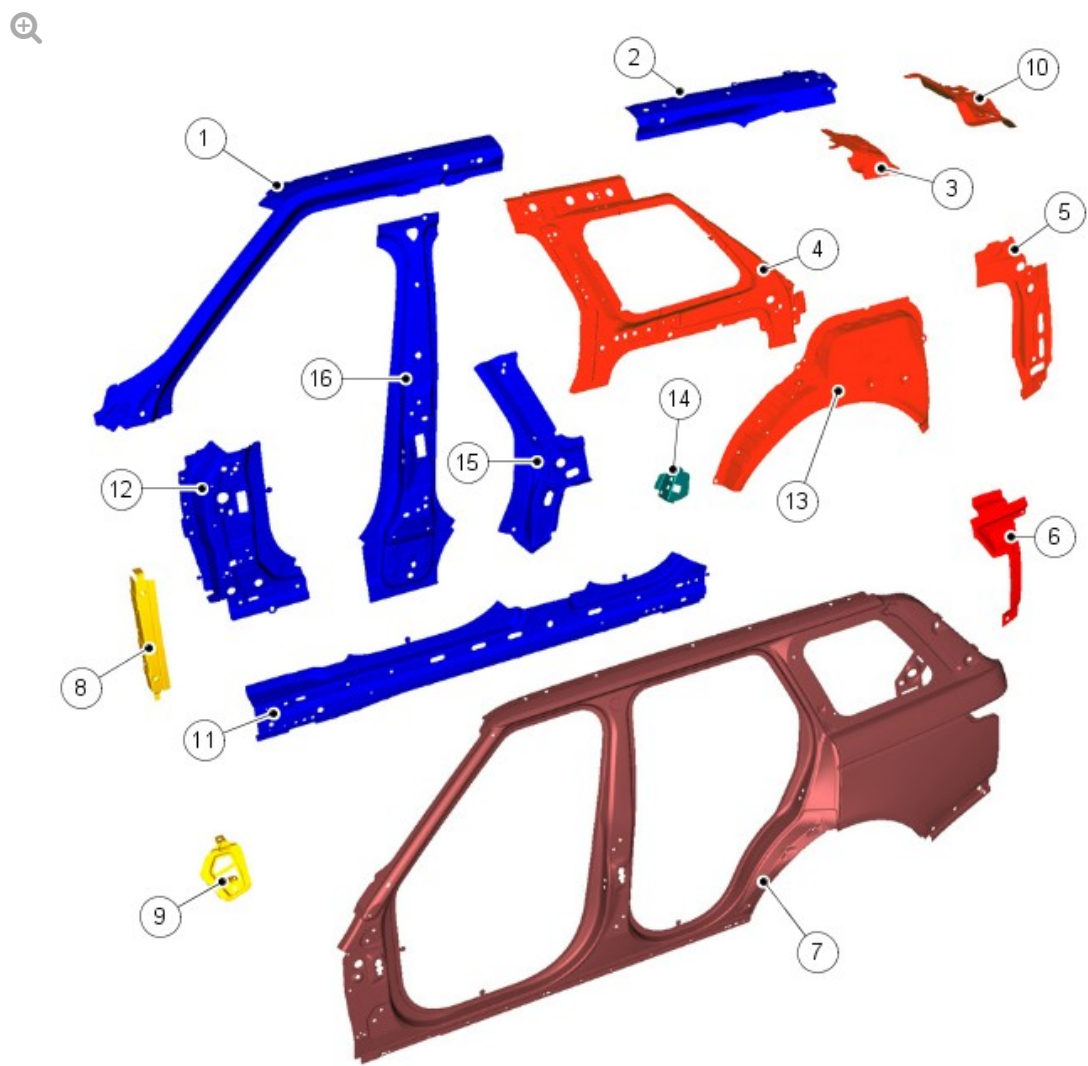
Side panels



E144638

ITEM	MATERIAL CODE	MATERIAL DESCRIPTION
1	6111T4	aluminum
2	6111T4	aluminum
3	5754NG	aluminum
4	6111T4	aluminum
5	5754NG	aluminum
6	5754NG	aluminum
7	6111T4	aluminum
8	6111T4	aluminum

Side panels - continued

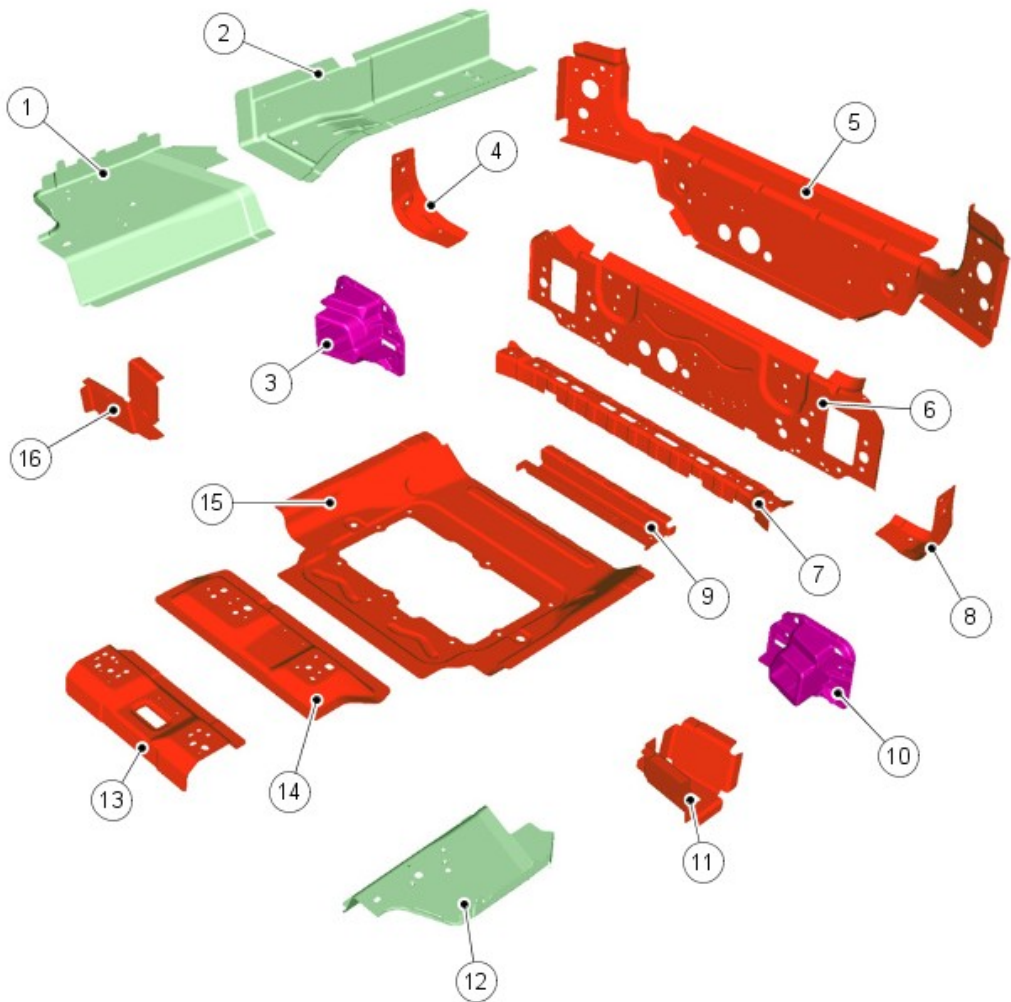


E144639

ITEM	MATERIAL CODE	MATERIAL DESCRIPTION
1	6111T4	aluminum
2	6111T4	aluminum
3	5754NG	aluminum
4	5754NG	aluminum
5	5754NG	aluminum
6	5754NG	aluminum
7	6111T4 + PFHT	aluminum
8	5182	aluminum
9	5182	Steel

10	5754NG	aluminum
11	6111T4	aluminum
12	6111T4	aluminum
13	5754NG	aluminum
14	MS1-3	aluminum
15	6111T4	aluminum
16	6111T4	aluminum

Rear end panels

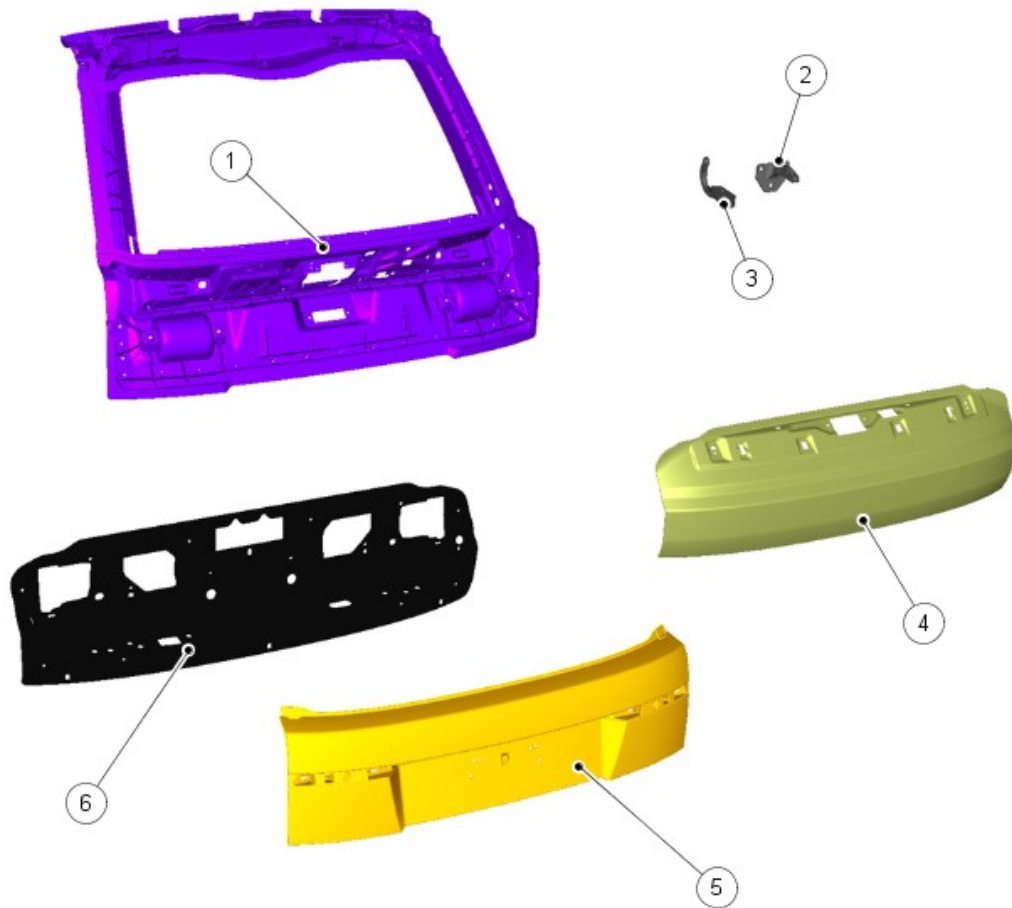


E144640

ITEM	MATERIAL CODE	MATERIAL DESCRIPTION

1	AC300 T61	aluminum
2	AC300 T61	aluminum
3	ADCHT	aluminum
4	5754NG	aluminum
5	5754NG	aluminum
6	5754NG	aluminum
7	5754NG	aluminum
8	5754NG	aluminum
9	5754NG	aluminum
10	ADCHT	aluminum
11	5754NG	aluminum
12	AC300 T61	aluminum
13	5754NG	aluminum
14	5754NG	aluminum
15	5754NG	aluminum
16	5754NG	aluminum

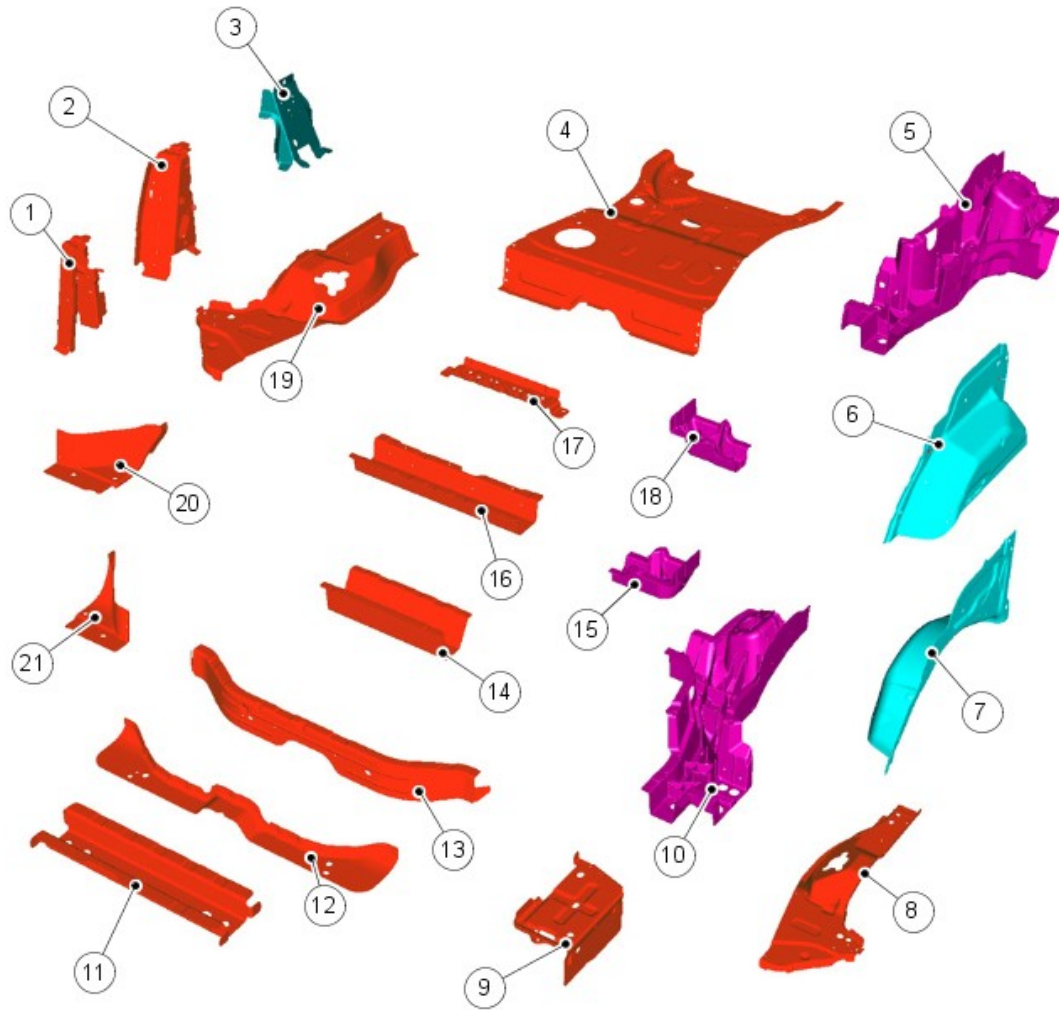
Rear end panels - continued



E144641

ITEM	MATERIAL CODE	MATERIAL DESCRIPTION
1	SMC	aluminum
2	-	Steel
3	-	Steel
4	AC170PX	aluminum
5	5182	aluminum
6	5754 NG	aluminum

Floor panels - rear

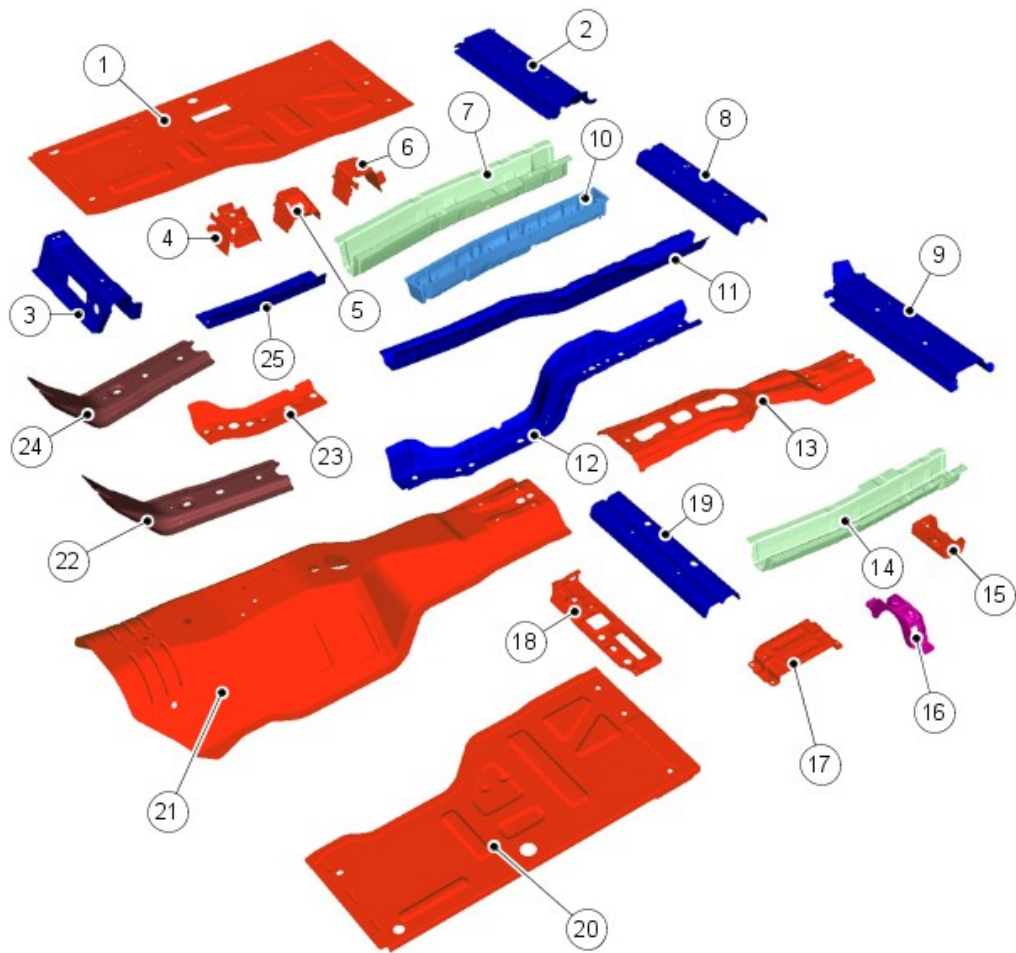


E144642

ITEM	MATERIAL CODE	MATERIAL DESCRIPTION
1	5754NG	aluminum
2	5754NG	aluminum
3	MS1-3	Steel
4	5754NG	aluminum
5	ADCHT	aluminum
6	5754NG	aluminum
7	5754NG	aluminum
8	5754NG	aluminum
9	5754NG	aluminum
10	ADCHT	aluminum

11	5754NG	aluminum
12	5754NG	aluminum
13	5754NG	aluminum
14	5754NG	aluminum
15	ADCHT	aluminum
16	5754NG	aluminum
17	5754NG	aluminum
18	ADCHT	aluminum
19	5754NG	aluminum
20	5754NG	aluminum
21	5754NG	Steel

Floor panels - center



E144643

ITEM	MATERIAL CODE	MATERIAL DESCRIPTION
1	5754NG	aluminum
2	6111T4	aluminum
3	6111T4	aluminum
4	5754NG	aluminum
5	5754NG	aluminum
6	5754NG	aluminum
7	AC300 T61	aluminum
8	6111T4	aluminum
9	6111T4	aluminum
10	PA66	aluminum

11	6111T4	aluminum
12	6111T4	aluminum
13	5754NG	aluminum
14	AC300 T61	aluminum
15	5754NG	aluminum
16	ADCHT	aluminum
17	5754NG	aluminum
18	5754NG	aluminum
19	6111T4	aluminum
20	5754NG	aluminum
21	5754NG	aluminum
22	6111T4 + PFHT	aluminum
23	5754NG	aluminum
24	6111T4 + PFHT	aluminum
25	6111T4	aluminum

GENERAL INFORMATION

CONTENTS

This document includes technical data and information for the all new Range Rover L405 aluminum range. The information contained within is valid at the time of production and incorporates:

- Panel replacement times
- Fixings
- Bonding
- Materials
- aluminum Welding

- Tolerances and gap specification
- Body alignment

In line with Land Rover continuous improvement program, information and data contained will be updated periodically.

All activities described within are based on the use of genuine Land Rover Parts, tools and approved materials.

PANEL REPLACEMENT TIMES

Panel replacement times are available for the following:

- Front end sheet metal repairs
For additional information, refer to: [Front End Sheet Metal](#) (501-27 Front End Sheet Metal Repairs, Description and Operation).
- Roof sheet metal repairs
For additional information, refer to: [Roof](#) (501-28 Roof Sheet Metal Repairs, Description and Operation).
- Side panel sheet metal repairs
For additional information, refer to: [Side Panel Sheet Metal](#) (501-29 Side Panel Sheet Metal Repairs, Description and Operation).
- Rear end sheet metal repairs
For additional information, refer to: [Rear End Sheet Metal](#) (501-30 Rear End Sheet Metal Repairs, Description and Operation).

FIXINGS

There are four groups of fixings used on the all new L405 aluminum range, other than conventional nuts and bolts. It is important that the correct procedures are followed for the removal and replacement of these fixings and, where applicable, the correct tools are used.

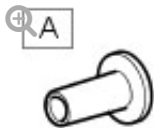
- Self-Piercing Rivet (SPR)
- Breakstem Fastener
- Countersunk Breakstem Fastener

- Torx Screw and Rivnut

SELF-PIERCING RIVET (SPR)

CAUTION:

Operating the SPR installation tool with no SPR or material in position will damage the tool die.



E74390

Self-Piercing Rivets are available in various sizes dependent on the size of the stack of panels to which they are fitted. A stack refers to the combined gauge of the panels being riveted. The following table identifies the sizes and part numbers available:

NOTE:

It is important to identify and select the correct size fixing by referencing this table prior to installation.

SPR TYPE NO.	LR PART NUMBER	RIVET LENGTH	RIVET APPLICATION FORCE	DIE DESCRIPTION	DIE (SPX / AVDEL PART NO)
1a	LR043550	5 mm	25 kN	Flat Bottom	78230- 050809
1b	LR043550	5 mm	30 kN	Flat Bottom	78230- 050809
2	LR043556	5 mm	45 kN	Pip	78230- 050088
3a	LR043551	6 mm	25 kN	Flat Bottom	78230- 050809

3b	LR043551	6 mm	30 kN	Flat Bottom	78230-050809
4a	LR043552	6.5 mm	35 kN	Pip	78230-050088
4b	LR043552	6.5 mm	30 kN	Flat Bottom	78230-050809
4c	LR043552	6.5 mm	30 kN	Pip	78230-050088
5a	LR043553	7 mm	30 kN	Flat Bottom	78230-050809
5b	LR043553	7 mm	40 kN	Pip	78230-050088
5c	LR043553	7 mm	35 kN	Flat Bottom	78230-050809
5d	LR043553	7 mm	50 kN	Flat Bottom	78230-050809
5e	LR043553	7 mm	40 kN	Flat Bottom	78230-050809
6	LR043559	7 mm	50 kN	Flat Bottom	78230-050809
7a	LR043554	8 mm	50 kN	Flat Bottom	78230-050809
7b	LR043554	8 mm	45 kN	Pip	78230-050088
7c	LR043554	8 mm	35 kN	Flat Bottom	78230-050809
7d	LR043554	8 mm	45 kN	Flat Bottom	78230-050809
7e	LR043554	8mm	50 kN	Pip	78230-050088
8a	LR043560	8 mm	40 kN	Flat Bottom	78230-050809
8b	LR043560	8 mm	50 kN	Flat Bottom	78230-050809
9	LR043555	9 mm	50 kN	Flat Bottom	78230-050809
10	LR043558	9 mm	50 kN	Flat Bottom	78230-

The two dies referred to in this table are: **"Pip die"** - Jaguar/Land Rover SPR repair kit, SPX/Avdel part number 78230-050088.



E149848

"Flat Bottom die"- Jaguar/Land Rover SPR repair kit, SPX/Avdel part number 78230-050809.



E149849

Removal

The ESN50 is the approved removal tool for Self Piercing Fixings. The ESN50 can be used with an optional, larger, 200mm C frame for improved access. Where tool access is impossible, drill out the remnant using a 5.3mm Cryobit drill bit. Remove all debris.

Installation

The ESN50 is the approved installation tool for Self Piercing Fixings. Where

the original panel is being refitted, the new fixings should be placed adjacent to the original. Only where the ESN50 cannot be used, or there is insufficient space on the panel, should the fixings be replaced with Breakstem Fastener Type As, unless the repair method dictates otherwise.



Using the ESN50

Make sure there is access for the ESN50 on both sides of the fixing to be removed/installed, including the optional C frame if required.

Removal: Align the plunger and anvil over the fixing and depress the trigger. Depressing the trigger will allow the plunger to act on the fixing and remove it from the panel.

NOTE:

Prior to replacing a self-piercing fixing, a test, using identical materials from waste panels, should be performed, to make sure that all settings are correct and an acceptable joint has been achieved.

Installation: Load new fixing into ESN50. Align the ESN50 over the fixing position and depress the trigger. As the trigger is depressed the ESN50 will clamp itself onto the work piece and install the fixing.

Removal of service SPR

Service SPR's will require the service anvil to have an internal diameter of 9mm, as shown in graphic E152490.



E152490

ITEM	DESCRIPTION
A	9mm

BREAKSTEM FASTENERS

Breakstem Fasteners are used in repair if the joint/stack cannot be accessed from both sides or require a greater load.

- Breakstem fasteners will cope with joint/stack sizes of 1.5mm to 10.8mm.
- A pre-drilled 6.5mm hole will be required before fitment.
- To identify the correct size and part number of structural breakstem fasteners to be used in repair always refer to TOPlx.

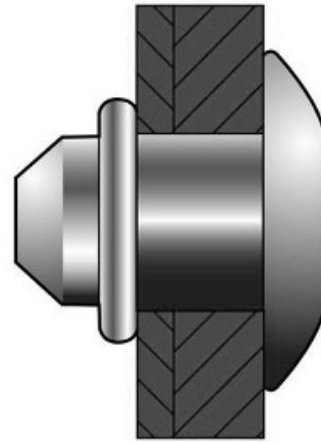
Where original rivet holes cannot be used replacement rivets should be positioned close to the original rivet location/hole location.

When riveting, the first location of the first replacement panel is close to the original rivet placement, not in the centre. The same process is followed by the next replacement, again not in the centre between the rivets.

BREAKSTEM FASTENER



E146646



Breakstem Fasteners have the following benefits:

- Exceptional shear and tensile strength
- Large blind side bearing area against the rear sheet
- Interference lock formed by a splined stem
- Spreads the tail bearing/clamp load on the rear sheet
- Provides strong, vibration resistant joints
- No loose stems

When the repair procedure requires the fitment of a breakstem fastener the stack size is identified within the repair procedures (millimetres). The table below identifies the correct breakstem fastener to the stack size.

NOTE:

When selecting a Breakstem Fastener (BSF) make sure that the BSF selected is at a midpoint between the stack sizes listed in the BSF table. For example a BSF 3.8 rivet in a repair procedure will require a LR044283 rivet which has a stack size of 2.8 to 4.8mm.

BREAKSTEM FASTENER TYPE	PART NUMBER	GRIP RANGE
BSF A	LR044284	1.5 mm to 3.5 mm

BSF B	LR044283	2.8 mm to 4.8 mm
BSF C	LR044286	4.8 mm to 6.8 mm
BSF D	LR043912	6.8 mm to 8.8 mm
BSF E	LR044287	6.8-7.5mm to 9.5mm
BSF F	LR044285	8.8 mm to 10.8 mm
BSF G	LR044288	Alu Countersunk
BSF H	LR043932	2.0 mm to 9.5mm

Removal

Breakstem Fastener used in production are removed by removing the centre of the fixing using a 4mm punch. Once the centre has been removed, drill out the remnant using a 6.5mm Cryobit drill bit. Remove all debris.

Aftermarket Breakstem Fastener are removed in the same way as Breakstem Fastener used in production.

Installation

The Genesis G4 is the approved installation tool for Breakstem Fastener . Prior to replacement make sure the fixing hole has been drilled to 6.5mm, using a Cryobit drill bit and all debris has been removed.



E72044

Using the Genesis G4

Make sure the fixing hole has been drilled to 6.5mm and all debris has been

removed. Check that you have the correct size Breakstem Fastener for replacement and insert it into the Genesis G4. Use the weight of the Genesis G4 to apply light pressure as the Breakstem Fastener is inserted into its hole. Depress the trigger which will tighten the Breakstem Fastener in the hole, the wasted stud will be ejected into the Genesis G4 once a pre-determined pressure is reached.

TORX SCREW AND RIVNUT



A Torx Screw and Rivnut are used where the fitment of a Breakstem Fastener Type A would give a poor cosmetic appearance. There are also occasions where the Torx Screw and Rivnut replaces a Breakstem Fastener Type B, where there is no access for the Genesis G4. The Rivnut acts as the thread.

DESCRIPTION	LAND ROVER SERVICE NUMBER
Rivnut (G)	TBC
Torx Screw (J)	TBC

Removal

The removal of the Torx Screw is carried out using a T30 Torx Driver. The Rivnut is not an original fixing.

Installation



E72042

The Torx Screw is installed using a T30 Torx Driver. The Rivnut is installed using the Wurth HES412 Rivet Nut Thread Setter, (Part No: 964948900).

ALUMINUM WELDING

Health and Safety

Refer to Land Rover Global Tools and Equipment Standards and Equipment Manufactures Manual.

Set Up/Equipment Check

Refer to equipment manufacturers operators' manual.

Weld Process

This section provides information on aftermarket aluminum welding for the all new L405 aluminum range. It covers the following areas:

- Equipment: Metal Inert Gas (MIG)
- Materials: Extruded and pressed aluminum sheet alloy.
- Weld Structure: Fusion welding process: Metal Inert Gas (MIG).
- Weld Procedures: Pre-weld, test weld, weld and post weld checks.
- Weld Variants: Lap weld, Butt weld, Plug weld.

Equipment

NOTE:

The Global Body Centre Standards have been published on the HUB, along with the personal protective equipment requirements, at the following locations:

- <https://jaguarhub.hub.tagworldwide.com/index.cfm?brand=42701>
- <https://landroverhub.hub.tagworldwide.com/index.cfm?brand=42701>

Only equipment that meets or exceeds the minimum welder standard in the Global Body Centre Tools and Equipment standards should be used in the repair of Land Rover products.

The accuracy of the approved equipment enables the delivery of one droplet of wire per pulse. This enables welding of incredibly thin parent material, i.e. 0.6mm sheet can be welded using a 1.2mm filler wire.

The approved MIG welder uses Direct Current, (DC) and does not produce High Frequency, (HF), at start up, (initial arc).

Welder Set Up

NOTE:

The user should always refer to the operators' manual for detailed instructions.

The following provides an overview of the set-up procedure:

- Health and Safety - read operators' manual
- Check correct power supply
- Check gas supply

- Check all equipment parts - safety check
- Attach power cables, connect to supply
- Purge gas bottle, attach
- Attach clamp

Approved equipment obtained from the Land Rover Equipment Program :

- Speedglas with Adflo face shield/weld mask
- Approved fume extractor
- Stainless Steel Brush
- Personal Protection Equipment (PPE)

Materials

Prior to any welding activity, it is necessary to identify the type of material to be welded. The all new L405 aluminum range is constructed from a number of aluminum alloys, each has different attributes that should be considered prior to the weld process.

Body Materials

The repair methods shown focus on the replacement of:

- Pressed aluminum Alloy sheet - 6xxx series - Used in skin-panels
- Extruded aluminum Alloy - 5754 - Used in structural panels

Weld Wire

The approved weld wires for these alloys are:

- 6xxx series - 4043 or 4047 filler wire
- 5754 - 5554 filler wire

Weld Structure

The approved aftermarket weld process is based on Fusion Welding:

- Metal Inert Gas (MIG) Welding

The approved MIG welder uses DC (direct current). The Electrode, (filler wire), is Positive Pole and the work piece is Negative Pole. The arc burns between a melting electrode, (which also acts as the filler wire) and the work piece. The shielding gas is Argon, Helium or a mixture of these, (MIG).

Successful aluminum welding is partly dependent on the removal of surface Oxidization. Oxide **MUST** be removed prior to welding. The oxide melts at a different temperature:

- aluminum melting temperature - approximately 660°C
- aluminum Oxide melting temperature - approximately 2040°C

WELD PROCEDURES

Pre-Weld

Prior to starting any weld procedure, the following safety precautions should always be implemented:

- Disconnect the vehicle battery
- Disconnect the generator electrical connectors
- Remove any ECM (engine control module)'s within 500mm of the weld area
- Remove the battery before carrying out welding work in its vicinity
- Utmost care must be taken when welding near the fuel tank or other components that contain fuel. If the tank filler neck or a fuel line must be detached to allow access for welding work, then the fuel tank must be drained and removed
- Adjacent vehicle parts and adjacent vehicles must be shielded against flying sparks and heat

The aluminum surfaces **MUST** be cleaned prior to welding. Cleaning will aid

penetration and avoid contamination. There are three steps to this process:

1. Clean surface:

To remove wax and any other contaminants use a chemical surface cleaner:

- Land Rover approved product: DuPont 3608S
- Alternatively, an Isopropyl based product

2. Remove Oxide

This can be achieved through a number of processes:

NOTE:

Items used to remove oxide should not be contaminated by use on a steel vehicle.

- Stainless Steel brush
- 80 Grit abrasive paper
- Non-Metallic scuff pad

3. Remove Oxide Dust

- Remove dust with a lint free cloth

NOTE:

Oxide builds up very quickly, therefore, perform steps 2 and 3 immediately prior to welding. If left for a period of time, steps 2 and 3 should be repeated.

MATERIALS

The materials listed in this section are those approved to be used on the all

new L405 aluminum range in conjunction with the repairs shown in this manual.

Approved Materials

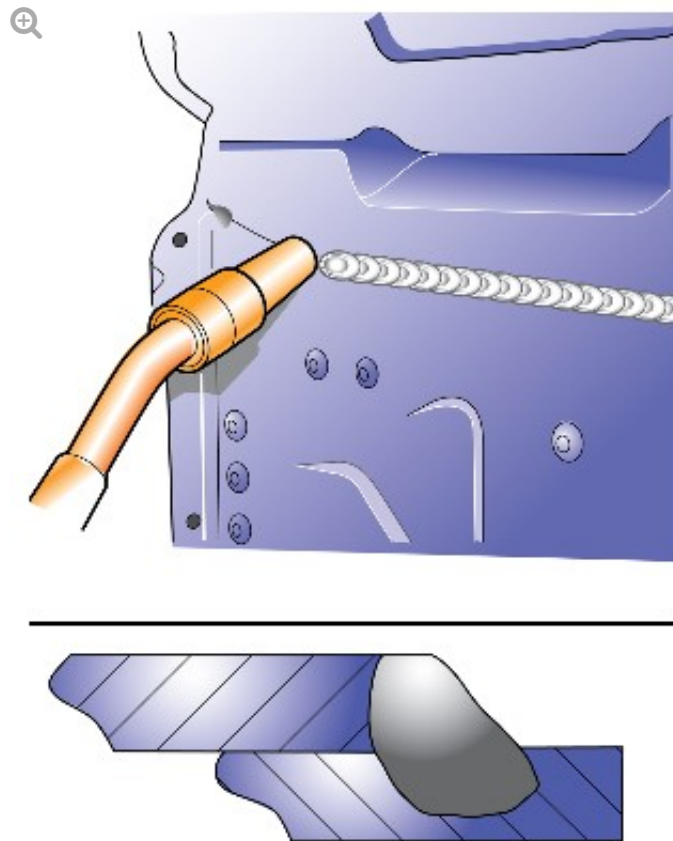
CONSUMABLE	MATERIAL	SUPPLIER
Weld Crack Penetrant	Rocol	Land Rover Equipment Program
Weld Crack Developer	Rocol	Land Rover Equipment Program
Weld Crack Cleaner	Rocol	Land Rover Equipment Program
Pyrosil coupling agent (primer)	SurAlink GP15	Land Rover Equipment Program
Pyrosil refill cartridge	Sura - Pyrosil	Land Rover Equipment Program
Bonding Adhesive	Structural Adhesive DP490	3M
Bonding Adhesive	Structural Adhesive 8115	3M
Cavity Foam	DM4330 Foam	3M
Semi Rigid Sealer	0893-2251	Wurth
Seam Sealer	Terostat 1K PUR 11272C (4500010)	Teroson
Cleaner/De-greaser	08984	3M
Cleaner/De-greaser	3608S	DuPont
MIG Weld Wire (5754)	5554 Filler Wire	Source Locally
MIG Weld Wire (6xxx)	4043 / 4047 Filler Wire	Source Locally
Abrasive Discs	Roloc Bristle Discs 07528	3M
Self Piercing Fixings (all variants)	Microcat	Land Rover Dealer Network
Breakstem Fasteners Fixings (all variants)	Microcat	Land Rover Dealer Network

WELD VARIANTS

The following identifies the three types of weld variants used in Category A panel replacement. It is not a guide to welding, as all Category A repairers will have staff who have attended and completed the Jaguar Land Rover aluminum welding accreditation.

This section highlights the key points for each weld variant.

Lap Weld

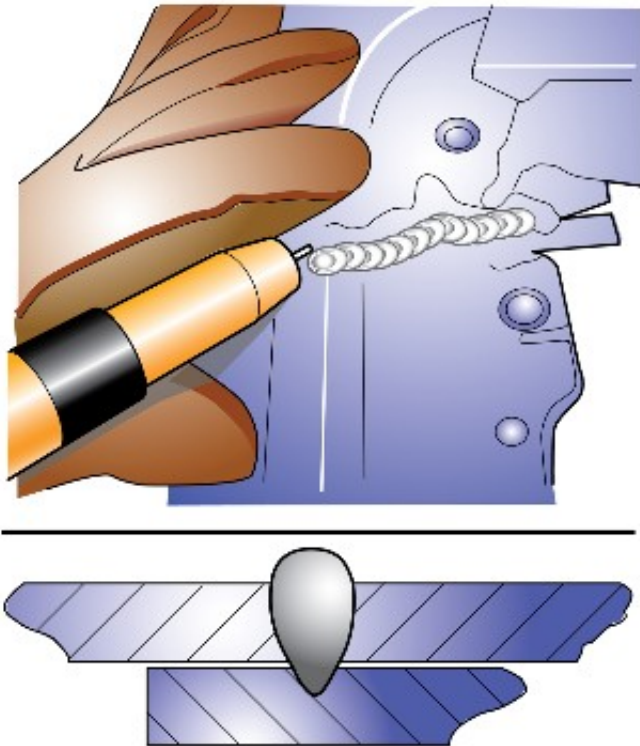


E141719

Key points:

- Run-In/Out distance of 5mm
- Minimum overall length of 20mm
- Must use Start/Finish weld tabs

Butt Weld



E141720

Key points:

- Run-In/Out distance of 5mm
- Minimum overall length of 20mm
- Must use Start/Finish weld tabs
- Backing plate required

Plug Weld



E141721

Key Points:

- Hole size 10mm
- Position of MIG nozzle - vertical

Backing Plates

NOTE:

Within certain repair procedures there is a requirement to carry out seam welding without the installation of welding backing plates. This is due to insufficient clearance between the two components. Please note that run on/run off tabs must still be used.

Backing plates are required for all butt-weld joints. Backing plate fabrication is made from waste material and have a width of 50mm. Securing backing plates is by either an interference fit, lap-weld, plug-weld or self-tapping

screws. Backing plate thickness is determined by work piece thickness, for example; (3mm work piece will require a 3mm backing plate). Backing plates are used to avoid excess penetration, dissipate the heat and supports the molten weld pool while cooling.

Weld Tabs

Weld tabs are manufactured from waste material and placed at the start and end of the effective weld length.

Test Weld

A test weld using identical coupons from waste panels should take place prior to working on the vehicle. The test piece is then visually checked and destructively tested to make sure all settings are correct and an acceptable weld has been achieved.

Prior to starting, the following should be checked:

- Gas - Correct for the type of job
- Filler Wire - Correct for the type of material to be welded

An effective weld should demonstrate the following qualities:

- All visible weld surfaces shall be clean, bright and of a uniform profile
- The weld seam should show uniform height and width over its entire length
- There shall be complete fusion between the surfaces of the work piece and the weld metal deposit

Correct level of penetration will be visible at the rear of the coupon as a fine continuous line.

Effective Weld Length

The effective weld length is the weld seam as described in the Body Repair Sections. The effective weld length does not include allowances for the run-

in/run-out, or termination defects, (start/stop), of the weld seam. It is permissible that the overall weld length is longer than detailed in the repair section, as the overall weld will include a minimum of 5mm at both the start and the stop of the weld seam, provided the function of the part is not affected, or the weld finishes on the edge of a panel.

POST WELD CHECKS

Weld inspections take the form of a visual examination, non-destructive and destructive testing.

Visual Examination

A visual examination of fusion welds should be carried out in accordance with the acceptance criteria detailed in the following Imperfections/Defect Levels table.

Imperfection/Defect Levels Table

DEFECT/IMPERFECTION TYPE	DETAILS	LIMITS
Burn-through	Burn holes	Not permitted
Seam offset, sides melted away	Incomplete fusion	Not permitted within effective weld length
Cracks	Any form of cracking is not permitted at any position along the entire length of the weld seam	Not permitted
Fused weld spatter		Limited acceptance. Not permitted on visible surfaces or in areas where functional performance of the part is affected, e.g. mating surfaces, sealing surfaces, etc. In such instances spatter is to be removed. All loosely adherent spatters must be removed
Visible ignition marks	Local melting of parent metal due to arc	Permitted provided functional performance of the component is not affected.
Open end crater	Reduces the cross	Not permitted

	sectional area of the weld seam	
Visual pores		Not permitted
Weld skip	Discontinuity/interruption in weld seam	Not permitted at any position along the entire length of the weld seam

Non Destructive

Dye penetrate testing **MUST** be used for detection of discontinuities, such as cracks, laps, folds, porosity and lack of fusion that are open to the surface of the material. Typical defects include start, (cold start/incomplete fusion) and stop, (crater cracking), defects within a fusion weld run. In addition to this lack of fusion/coalescence at the weld toe, solidification cracks in the weld bead may also be detected.

Non Destructive Crack Inspection Process

Use the product as listed in the Approved Materials Section, this product is supplied within the Category A tool kit. The product is an aerosol applied dye system. It is designed to penetrate the finest cracks and flaws to facilitate detection, the system includes: Cleaner, Penetrant and Developer.

The process is as follows:

- 1. Use the cleaner to de-grease/clean the test area, then wipe with a lint free cloth.
- 2. When the surface is completely dry, apply the penetrant. Cover the test area and allow a minimum contact time of 10 minutes.
- 3. Remove excess penetrant from the surface with a lint free cloth wetted with the cleaner.
- 4. Apply a THIN film of the developer and leave for a minimum of 10 minutes to draw up the retained dye from flaws or cracks.
- 5. Suspect areas should then be examined under natural or electric light for signs of flaws and/or cracks. Cracks will show up as lines whilst

porosity will appear as pin holes.

Destructive Testing

Each test weld should be pulled apart to check the quality and penetration of the weld.